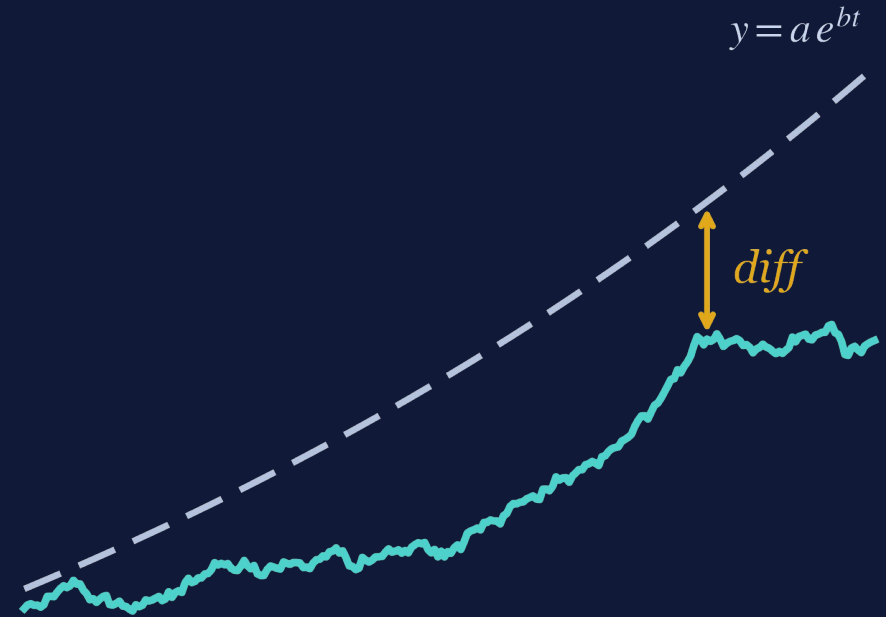


Risk-Adjusted Performance Analysis Based on Data

Five classic ratios, one new metric — $\text{diff} \times r \times b$ — tested across 100 ETFs and 9 market regimes.



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ABSTRACT

One new metric against five classics

The project examines the **Sharpe, Treynor, Calmar, Sortino** and **Information** ratios, and proposes a new metric derived from an exponential regression on the asset price.

$$\text{Score} = \text{diff} \times r \times b$$

Across 100 ETFs and 9 market regimes, the new metric acts as a **mean-reversion signal**, while the classic ratios behave as **momentum signals**.

Combining them according to the market regime outperforms each one individually.



Price deviation

Distance of the current price from the trend line



Fit quality

Correlation coefficient — quality of fit to the trend



Growth slope

Daily growth rate of the fitted trend

The five classic ratios

Each ratio highlights a different aspect of risk — and all five reward strong recent risk-adjusted returns.

Sharpe • 1966

$$\frac{R_p - R_f}{\sigma_p}$$

Excess return per unit of total volatility

Sortino • 1994

$$\frac{R_p - R_f}{\sigma_{\text{down}}}$$

Excess return per unit of downside volatility only

Treynor • 1965

$$\frac{R_p - R_f}{\beta_p}$$

Excess return per unit of systematic risk (β)

Information • 1989

$$\frac{R_p - R_b}{\text{TE}}$$

Consistency of outperformance versus benchmark

Calmar • 1991

$$\frac{R_{\text{ann}}}{|\text{MaxDD}|}$$

Annual return relative to maximum drawdown

$$\text{Score} = \text{diff} \times r \times b$$

We fit an exponential model to the historical price — an OLS regression on log-price:

$$y = a \cdot e^{bt}$$

r

Fit quality

Pearson correlation between log-price and time ($|r| \leq 1$)

b

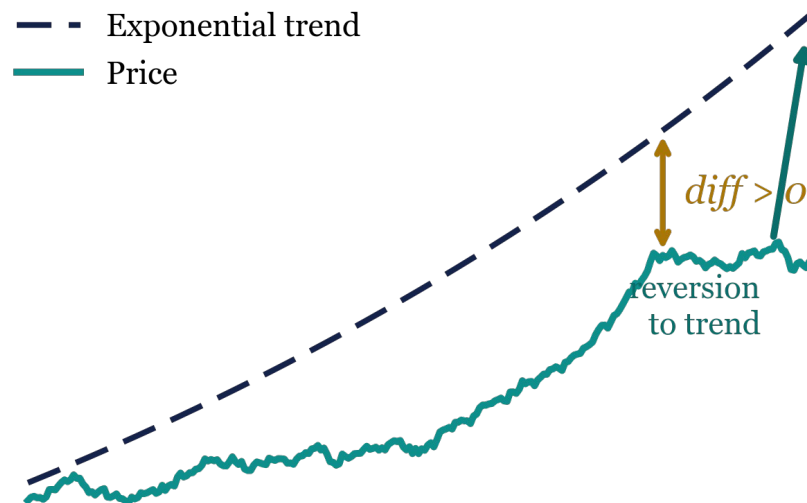
Growth slope

Daily growth rate (log-growth rate)

diff

Price deviation

$\text{ExpReg}(\text{last}) - \text{Price}(\text{last})$: signed distance from the trend



Price below the trend + positive trend → positive score → **predicted UP** (mean reversion).

Experiment design

100

exchange-traded funds

180

configurations scanned

9

market regimes

5 × 4

training × test windows

1

Training window

Compute all 6 metrics on each of the 100 funds.

2

Test window

Observe direction and realize returns.

3

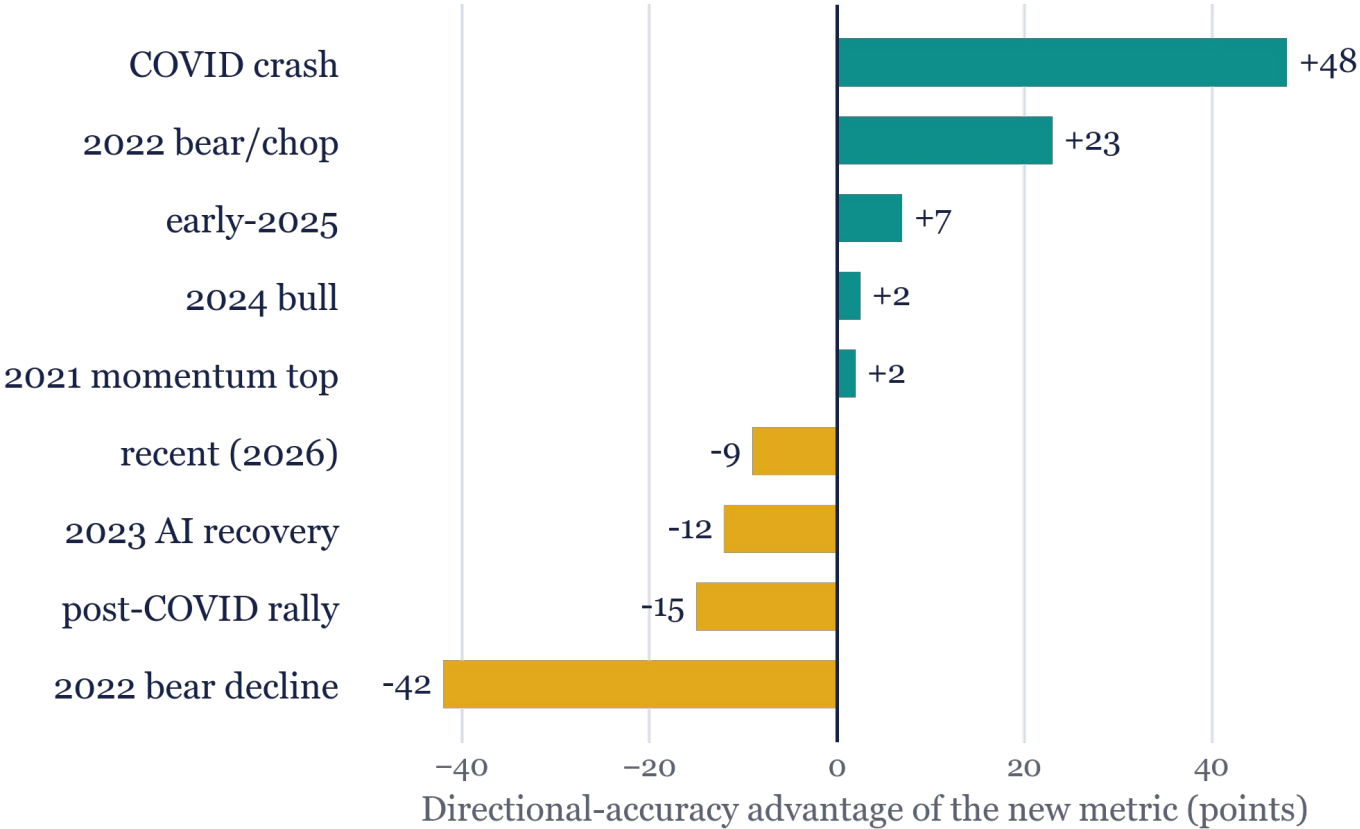
Evaluation

Directional accuracy and long/short portfolio return.

Each metric is converted into a directional forecast: **a positive score predicts an increase, a negative score predicts a decrease.**

In addition, a dollar-neutral long/short portfolio is constructed — 10 assets on each side, weights proportional to $|\text{score}|$, with a strict cap on the short side.

A regime-dependent edge



The new metric wins
Reversals and volatile markets

Classic ratios win
Persistent trends

Key takeaway
The pattern is not random — it is the signature of a mean-reversion signal versus a momentum signal.

When reversion pays

CASE A • COVID CRASH REBOUND (2020)



96.9% vs **9.2%**

directional accuracy — new metric vs classic ratios

The training window ended near the March 2020 bottom. Almost all ETFs sat below their exponential trend → large positive diff → predicted UP. The classic ratios, depressed by catastrophic returns, forecast continued decline.

Long/short return: +10.9% vs -8.8%

CASE B • 2022 CHOPPY BEAR TAPE (Q4)



90.0% vs **22.6%**

directional accuracy — new metric vs classic ratios

A market with no clear direction — mean-reverting. Extended names below trend snapped back; names above it faded. The advantage is not unique to a single crash.

Long/short return: +8.1% vs +2.4%

When the trend persists

CASE C • POST-COVID MOMENTUM RALLY



2.0% vs **93.9%**

directional accuracy — new metric vs classic ratios

End of 2020 — the market had risen for months. Most ETFs were above their trend → negative diff → predicted a pullback that never came. Momentum continued; the classic ratios were right in almost every case.

Long/short return: - vs **momentum carried on**

CASE D • 2022 SUSTAINED DECLINE



8.1% vs **79.0%**

directional accuracy — new metric vs classic ratios

A prolonged, directional decline. Names below trend kept falling — the mean-reversion signal “caught a falling knife.”

Long/short return: **-6.4%** vs **+7.8%**

When the trend persists, mean-reversion forecasting is systematically wrong — the mirror image of Case A.

Two complementary signal types

The new metric

MEAN REVERSION

Positive when price is below a stable trend. Predicts a return to trend. Wins in reversals and volatile markets.

The classic ratios

MOMENTUM

Large after strong risk-adjusted returns. Predict continuation. Win in sustained trends.

The literature (Jegadeesh & Titman, 1993; Daniel & Moskowitz, 2016) documents “momentum crashes” at reversals — Case A is a textbook example.

Takeaway: a regime-aware combination — switching between the two signal types according to volatility, trend strength, or the size of diff — is expected to outperform either family on its own.

Complementary, not competing

1

diff × r × b is a mean-reversion signal — not a replacement for the classic ratios.

2

The classic ratios behave as momentum signals — direct empirical evidence.

3

The two families are complementary — each wins where the other fails.

4

Regime-based combination outperforms any single metric on its own.

NEXT STEP

Walk-forward backtest

Roll the training window day by day, recalculating metrics and rebalancing portfolios into a continuous equity curve — testing:

- whether the regime-dependent edge survives sequential rebalancing
- regime filters (volatility, trend strength, $|diff|$) cycle by cycle
- transaction costs, turnover, and short-borrow constraints